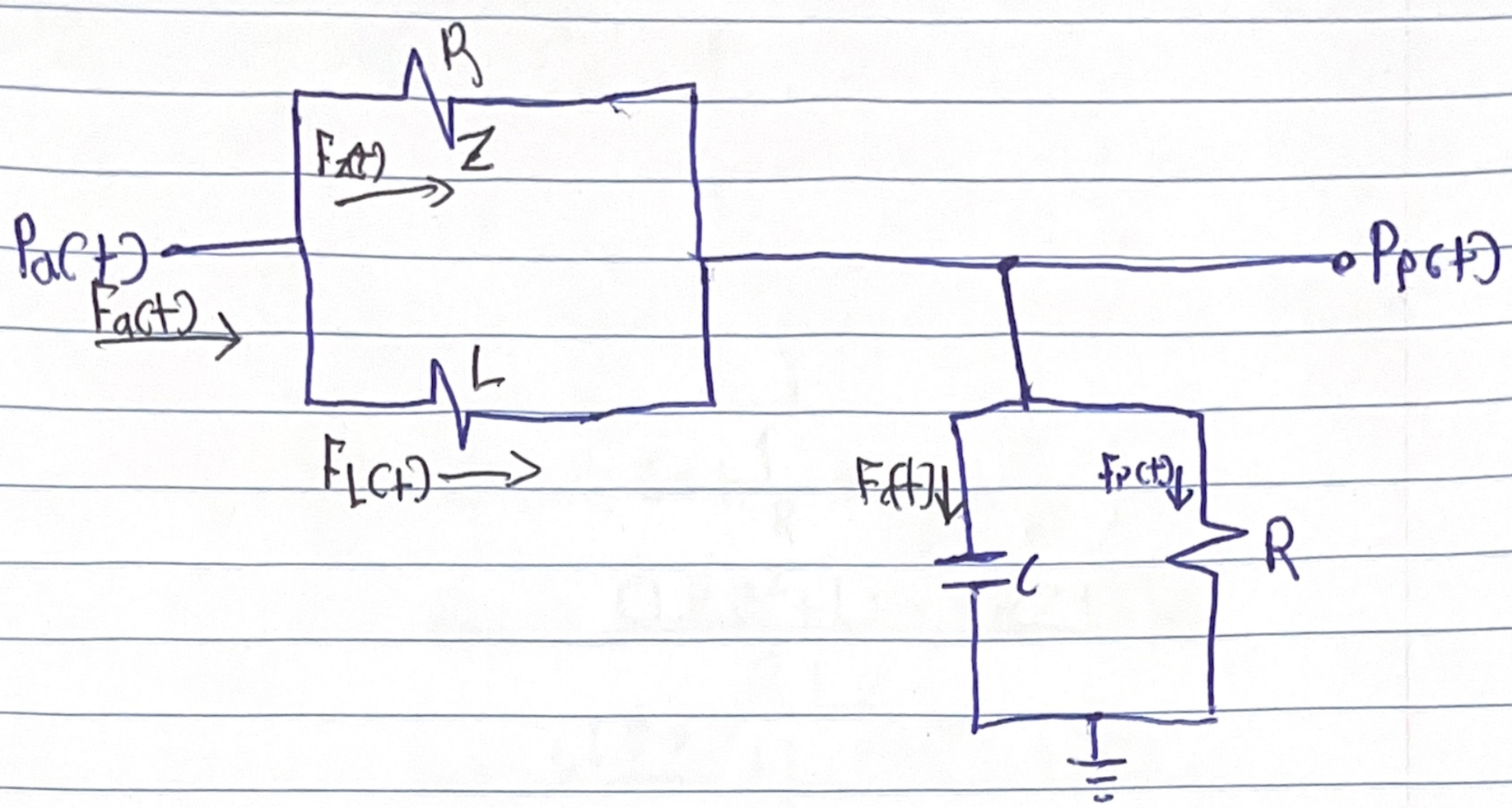


Función de transferencia

Ecuaciones principales

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int (P_a(t) - P_p(t))dt$$

$$F_c(t) = C\frac{dP_p(t)}{dt}$$

$$F_R(t) = \frac{P_p(t)}{R}$$

Transformada de Laplace

$$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$$

$$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$$

$$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$$

$$F_c(s) = C_0 P_p(s)$$

$$F_R(s) = \frac{P_p(s)}{R}$$

Procedimiento algebraico

$$F_Z(s) + F_L(s) = F_C(s) + F_R(s)$$

$$\frac{P_a(s) - P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{Ls} = Cs P_p(s) + \frac{P_p(s)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = Cs\, p_p(s) \frac{P_p(s)}{R}$$

$$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(Cs + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)$$

$$P_a(s) \frac{Ls+Z}{LZs} = P_p(s) \frac{CLRZs^2 + LZs + RZ + RLs}{RLZs}$$

$$P_a(s) \frac{Ls+Z}{LZs} = p_p(s) \frac{CLRZs^2 + L(C+R)s + RZ}{RLZs}$$

$$\frac{\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}}}{}$$

$$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(CZ+R)s + RZ} = \text{Resultado}$$

Estabilidad del sistema en lazo abierto

$$CLRZs^2 + L(CZ+R)s + RZ = 0$$

$a = CLRZ$

$b = L(Z+R)$

$c = RZ$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-L(Z+R) \pm \sqrt{L^2(C+R)^2 - 4CLR^2Z^2}}{2CLRZ}$$

Hipotenso

$Z = 0.010$

$C = 1.500$

$R = 1.000$

$L = 0.001$

$\lambda_1 = -12.067$

$\lambda_2 = -55.271$

Presenta una respuesta estable y sobreamortiguada a un escalon unitario de entrada

Normotenso

Z = 0.010
C = 3.500
R = 1.000
L = 0.001

$\lambda_1 = -14.428 + 8.805i$
$\lambda_2 = -14.428 + 8.805i$

Presenta una respuesta estable y subamortiguada a un escalon unitario

Hipertenso

Z = 0.010
C = 1.000
R = 1.000
L = 0.001

$\lambda_1 = -11.126$
$\lambda_2 = -89.873$

Presenta una respuesta estable y sobreamortiguada a un escalon unitario.

Error en esta estacionario

$$e(s) = \lim_{s \to 0} Pa(s)\left(1 - \frac{Pp(s)}{Pa(s)}\right) = e(s) = \lim_{s \to 0}\left(1 - \frac{LRs + RZ}{CLRs^2 + (LCZ+R)s + RZ}\right) = 0$$

$e(t) = 0V$

Modelo de ecuaciones integro-diferenciales

$$Fz(t) + FL(t) = Fc(t) + FR(t)$$

$$\frac{Pa(t) - Pp(t)}{Z} + \frac{1}{L}\int [Pa(t) - Pp(t)]dt = C\frac{dPp(t)}{dt} + \frac{Pp(t)}{Z}$$

$$\left(\frac{1}{R} + \frac{1}{Z}\right) Pp(t) = \frac{Pa(t)}{Z} + \frac{1}{L}\int (Pa(t) - Pp(t)dt - C\frac{dPp(t)}{dt}$$

$$Ppc(t) = \left[\frac{Pa(t)}{Z} + \frac{1}{L}\int (Pa(t)dt - C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$